

Mani Bharathi R

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PROFESSIONAL SUMMARY

Results-driven Design Control and Quality Engineer with over 5 years of experience in medical device regulatory compliance, software lifecycle management, and technical documentation under IEC 62304, ISO 13485, and EU MDR 2017/745 frameworks. Proven expertise in SaMD, MRI software systems, and PLM process management, ensuring design integrity, risk mitigation, and global regulatory readiness (FDA/CE). Skilled in collaborating with cross-functional teams to drive design control excellence, manage defect resolution, and maintain traceability and compliance across the product lifecycle.

CORE COMPETENCIES

- Design Control & DHF Remediation
- Software Life Cycle Compliance (IEC 62304)
- ISO 13485 & ISO 14971 Implementation
- EU MDR & FDA Submission Support
- Risk Management & Hazard Analysis
- Verification & Validation (V&V) Documentation
- CAPA & Root Cause Analysis (RCA)
- Defect Tracking (IBM ClearQuest)
- PLM & Configuration Management (Windchill, TcU and SAP)
- Technical Documentation (STED and GSPR)
- Audit Preparation & Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

TATA Elxsi, Pune

Senior Engineer | Feb 2024 – Present

- Ensured IEC 62304 compliance for MRI system software, managing full life cycle phases — design, development, verification, validation, and maintenance.
- Collaborated with cross-functional teams to maintain robust Design Control documentation per ISO 13485 and Class C software requirements.
- Reviewed and approved Software Design, Requirement, and Verification deliverables, ensuring bidirectional traceability.
- Utilized IBM Rational ClearQuest for software defect tracking and closure.
- Supported Technical Documentation (STED) for CE and FDA submissions.
- Maintained Configuration and Change Control through Windchill PLM.
- Led RCA and CAPA initiatives to enhance process quality.

HCL Technologies, Madurai

Member Technical Staff | Sep 2021 – Feb 2024

- Remediated Verification Documents for Biocompatibility and E&L testing per ISO 10993 standards.
- Created and remediated Design History Files (DHF) addressing design inputs, outputs, and V&V gaps.
- Updated Risk Management Files (RMF) in alignment with ISO 14971 and MDR requirements.
- Developed TÜV Notified Body checklists for Biocompatibility compliance under EU MDR 2017/745.
- Served as Configuration Controller ensuring version control and QMS record accuracy under ISO 13485.

ZF Wabco India Pvt. Ltd., Chennai

Graduate Engineering Trainee | Aug 2020 – Aug 2021

- Operated and maintained VMC, Lathe, and Grinding machines as part of manufacturing operations.
- Conducted quality checks and dimensional inspections on crankshaft components.
- Supported process optimization initiatives to reduce quality rejections.

EDUCATION

Bachelor of Engineering (Mechanical Engineering)
Anna University, Chennai | 2020

TOOLS & APPLICATIONS

- IBM Rational ClearQuest – Defect & Change Management
- Windchill & TcU – Configuration & Documentation Control
- SAP – Drawing Review, Material Traceability
- MS Office & SharePoint – Documentation & Review Management